

Microbial Culture Research

METHODOLOGY

Sample Collection

Sterile containers were used to collect yogurt samples and store them at 4°C until further processing. The samples were serially diluted in sterile 0.85% saline solution.

Isolation of *Lactobacillus*

a) Media preparation:

MRS agar was prepared by mixing 7g of MRS powder in 100ml distilled water. This mixture was agitated using a magnetic stirrer and then both the media and petri plates were sterilized in the autoclave for 30 minutes to remove contamination. After autoclaving, media was poured into petri plates and was solidified by placing the plates in a laminar flow hood for 15 to 20 minutes.

b) Inoculation of *Lactobacillus*

MRS plates were labelled properly to indicate different levels of dilutions. Using a micropipette, the diluted yogurt samples were taken and carefully spread on the plates using a glass spreader for even spreading. All steps were performed under sterile conditions to prevent contamination.

c) Incubation under aerobic conditions

Plates were placed in an incubator at 30°C for 24–48 hours.

d) Incubation under anaerobic conditions

Plates were also inoculated under anerobic conditions using candle jar method. Silica gel and candles were used to absorb moisture and create anaerobic environment in the jar and then was sealed by using Vaseline to create airtight environment and prevent oxygen infiltration.

Sub-culturing:

After incubation, growth appeared. Colonies were observed carefully. *Lactobacillus* appeared as small, creamy-white, round colonies and was inoculated on fresh MRS plates using inoculating loop. Plates were inoculated under both aerobic and anerobic environment by in incubator and candle jar for 24 hours. Sub-culturing was repeated twice to obtain pure colonies.

Identification of *Lactobacillus*

Gram Staining

Gram staining is a differential staining technique used to classify bacteria as Gram-positive or Gram-negative based on the structure of their cell walls. A smear of the *Lactobacillus* culture was prepared on a glass slide and heat-fixed. The slide was stained with crystal violet for one minute, then rinsed and treated with iodine solution for another minute. After that, the slide was decolorized using alcohol or acetone for a few seconds and immediately rinsed. Finally, it was counterstained with safranin for 30 seconds and then rinsed with water. The slide was dried and observed under a microscope.

RESULTS

Isolation of *Lactobacillus* from Yogurt Samples

Home made yogurt samples were used as a source of lactic acid bacteria (*lactobacillus*). Samples diluted with distilled water were inoculated on MRS plates followed by incubation under both aerobic and anerobic conditions in incubator and candle jar at 37 °C for 24 hours.

Plates incubated in anaerobic jar showed less growth compared to ones in incubator.

Colonies were sub cultures to obtain pure culture.

Colonies appeared on all plates were creamy white and round having gleaming surfaces after 24 hours which is morphological characteristic of *lactobacillus* bacteria.



Fig 1: Vacuum desiccator setup used for the isolation of *Lactobacillus* under anaerobic conditions.

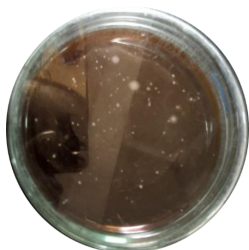


Fig.2: Primary culture of *Lactobacillus* obtained from anaerobic conditions.

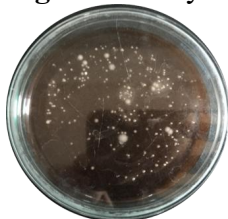


Fig3: Primary culture of *Lactobacillus* obtained from aerobic conditions.

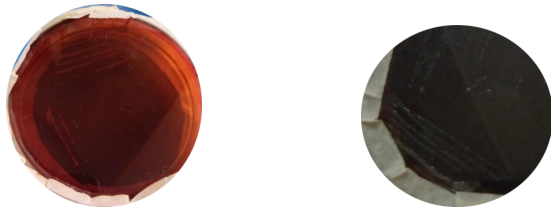


Fig.4: a) Secondary culture of *Lactobacillus* grown under aerobic conditions. b) Secondary culture of *Lactobacillus* grown under anaerobic conditions.



Fig5: Pure *Lactobacillus* colonies isolated

Gram Staining Results

Examination of gram stains under microscope indicated that isolates were gram positive arranged in short chains.

Cells appeared as purple violet stained under microscopic examination supporting the presences of *lactobacillus*.

No gram-negative isolates were detected.

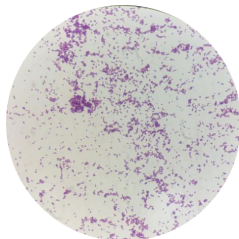


Fig6: Gram staining showing Gram-positive (purple–violet) *Lactobacillus* cells.

CONCLUSION

In this study *Lactobacillus* strain were obtained from yogurt sample, examined their morphological characterization, and thereafter tested their ability to produce bacteriocins in MRS media. Both MRS agar and broth worked as an effective media that provide suitable selective growth of lactic acid bacteria that facilitate the recovery of pure cultures. Microscopic analysis using Gram staining showed Gram-positive, rod-shaped organisms which is the characteristic identification of *Lactobacillus*.